a

Input Data

Sequence

5′ A A C G T A A C T 3′

101 nt

score

PhyloP Conservation Signal

Input Feature Encoding

One-Hot

A A C G T A A C T

5-Channel Feature

Multi-Scale Motif & Context

3

5

7

Concatenate

SE attention

1×1 fusion

Bidirectional GRU

LayerNorm

Dropout

Dual Output Heads

Profile Head

Binding Profile Logits

Softmax × exp(z)

Predicted RBP Binding Profile

Count Head

z

Log Total Count

b

Kernel size

size = 3

size = 5

size = 7

Concatenate

SE attention

Global mean pooling

1×1 96→24 · ReLU

1×1 24→96 · Sigmoid

1×1 fusion · 96→96

1×1 projection · 5→96

c

Bidirectional GRU

1×1 • 96→64

Concat · 128

1×1 96→128

LayerNorm

Dropout

Profile Head • 1×1 conv

128→64

64→32

32→1

Binding Profile Logits

Count Head

Global Mean Pooling

128→32

32→1

Log Total Count

d

Ground Truth

Predictions

Profile Shape Loss

Squared-CDF Loss

Count Loss

α

β

γ

$\alpha L_{Shape} + \beta L_{Jitter} + \gamma L_{Count}$

Total Loss

e

Predicted profile

Select high-IG windows

GUAACAGCUGUUGACG

GUAGCCGCUGUUGACG

GUACCGGCGUUCUGGA

GUAACACGCGUCUGGG

enriched motif segments

TF-MoDISCO

CWM

PPM

Motif Logos

f

Ground Truth

Predictions

Identify important motif-centered bases

CAUGCACAGAU

Enriched motif segments

CAUGCUCAGAU

Re-run prediction

reference

mutated

motif region

mutation site

Δ mutated